

Gustavo Valdivia-Mera

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Ph.D. Student in Physics, University of Houston
Department of Physics and Texas Center for Superconductivity

ORCID | INSPIRE-HEP

EDUCATION

- **University of Houston, Houston, Texas, USA** 2023–Present
Ph.D. in Physics *GPA: 4.0/4.0*
 - **Basic Core:** Methods of Mathematical Physics, Advanced Mechanics, Quantum Mechanics I–II, Electrodynamics, Statistical Physics.
 - **Advanced Core/Electives:** Quantum Field Theory I–II, Advanced Statistical Mechanics, General Relativity and Cosmology, Quantum Information.
- **ICTP-EAIFR, Kigali, Rwanda** 2020–2022
M.Sc. in High Energy Physics *Ranked 1st in cohort (HEP section)*
 - Thesis: *Bounds on eV -scale sterile neutrinos from neutrinoless double-beta decay.*
 - Completed the ICTP Diploma Programme in High Energy Physics during the M.Sc. program.
- **Universidad Nacional Mayor de San Marcos, Lima, Peru** 2018–2019
Professional Degree in Physics *Distinction: Outstanding*
 - Thesis: *On the Underlying Relation Between Quantum Entanglement and Wormholes.*
- **Universidad Nacional Mayor de San Marcos, Lima, Peru** 2012–2017
B.Sc. in Physics *Ranked 1st in cohort*

AWARDS AND HONORS

- La Guerrera - The Elva Morua de Rodriguez Scholarship, University of Houston 2026–2027
- Lydia Mendoza Fellowship, University of Houston 2026–2027
- Student Ambassador, American Physical Society (APS) 2026–2027
- Lydia Mendoza Fellowship, University of Houston 2024–2026
- Presidential Fellowship, University of Houston 2023–2025
- 1st Place, Quantum Computing Fall Fest, University of Houston 2023
- Research Scholarship, UNMSM Vice-Chancellor's Office 2018

RESEARCH EXPERIENCE

- **Hosur Group, University of Houston** 2026–Present
Research Assistant
 - Study theoretical problems in strongly correlated quantum many-body physics, topological quantum matter, and anomalous transport.
- **Ordóñez Lab, University of Houston** 2023–2025
Research Assistant
 - Investigated the emergence of thermality and information-theoretic structures in causally restricted spacetime domains, including causal diamond horizons and black holes.
- **Theoretical Physics Group, UNMSM** 2015–2019
Undergraduate Research Affiliate
 - Worked on the AdS/CFT correspondence, black hole thermodynamics, and the relation between entanglement and spacetime geometry.

PUBLICATIONS

- **Area-Information Trade-Offs in Acceleration Radiation from Atoms Falling into Black Holes**
Yusef Maleki, Gustavo Valdivia-Mera, Carlos R. Ordóñez, Horacio E. Camblong, and Marlan O. Scully
Physical Review D. 114, 065017 (2026). [Journal](#) | [arXiv](#)
- **Acceleration radiation and HBAR thermodynamics for atoms falling into a BTZ black hole: A CQM quantum-optics approach**
H. E. Camblong, A. Chakraborty, C. R. Ordóñez, M. O. Scully, G. Valdivia-Mera, and H. Wang
arXiv preprint. arXiv:2607.14281 (2026). [arXiv](#)
- **The Helical SYK Model and Emergent Infrared Integrability**
Gustavo Valdivia-Mera, Bhavay Tyagi, Eric R. Bittner, and Pavan Hosur
arXiv preprint. arXiv:2606.17161 (2026). [arXiv](#)

- **Thermal nature of the causal diamond horizon: A hidden property of the inertial propagator**
Nada Eissa, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Physical Review D. 112, 125008 (2025). [Journal](#) | [arXiv](#)
- **Horizon brightened acceleration radiation entropy in causal diamond geometry: A near-horizon perspective**
Nada Eissa, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Physical Review D. 112, 085030 (2025). [Journal](#) | [arXiv](#)
- **Path integral derivation of the thermofield double state in causal diamonds**
Abhijit Chakraborty, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Classical and Quantum Gravity. 42, 025015 (2025). [Journal](#) | [arXiv](#)
- **On the Unruh effect and the thermofield double state**
Gustavo Valdivia-Mera
International Journal of Modern Physics D. 34, 2530002 (2025). [Journal](#) | [arXiv](#)

SELECTED PRESENTATIONS

- **Strange Metals and Black Holes: A Holographic Correspondence** *March 2026*
Research Day, Department of Physics, University of Houston, Houston, Texas, USA
- **Thermal Nature of the Causal Diamond Horizon** *February 2026*
TINKUY 2026: “Un Panorama de la Física Teórica en el Perú”, Lima, Peru
- **Towards HBAR in the causal diamond near horizon** *July 2025*
IQSE Summer School on Quantum Science, Texas A&M University, Casper, WY
- **Near-horizon (conformal) aspects of black holes and the universality of HBAR entropy** *April 2025*
AMO/IQSE Seminar, Texas A&M University, College Station, TX
- **Path integral derivation of the thermofield double state in causal diamonds** *March 2025*
APS Global Physics Summit, Anaheim, CA
- **Quantum Algorithms I & II** *August 2023*
Quantum Scholars Program, QuantumQuipu, online
- **Bounds on eV-scale sterile neutrinos from neutrinoless double-beta decay** *June 2023*
Summer School on Particle Physics, ICTP, Trieste, Italy

TEACHING EXPERIENCE

- **University of Houston, Houston, Texas, USA** *2023–2025*
Teaching Assistant
 - Quantum Mechanics, undergraduate course *Fall 2025*
 - Quantum Mechanics II, graduate course *Spring 2025*
 - Physics Laboratory, undergraduate course *Fall 2023*
- **Universidad Privada del Norte (UPN), Lima, Peru** *2020–2025*
Teaching Assistant, Physics
- **TECSUP, Lima, Peru** *2018*
Instructor, General Physics
- **Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru** *2017–2018*
Teaching Assistant, Mathematics

TECHNICAL SKILLS

Python; Fortran; Mathematica; L^AT_EX.